



KARPAGAM
ACADEMY OF HIGHER EDUCATION
(Deemed to be University)
(Established Under Section 3 of UGC Act, 1956)
Accredited with A+ Grade by NAAC in the Second Cycle

HACKSPRINT 2.0



Team Name: Vector

Domain: Open Innovation

Category: Hardware

Project Title: RESCUE-ZERO: A No-Network Emergency Location System



Challenge Statement



No mobile network or power, north Chennai cries for help

Not one MLA or GCC official visited them in the last three days, says a resident

Updated - December 07, 2023 02:40 am IST - Chennai

THE HINDU BUREAU

THE TIMES OF INDIA

THIS STORY IS FROM DECEMBER 4, 2015

With limited ration, no power or internet access, Chennai still in deep water

Sivakumar B / TNN / Updated: Dec 4, 2015, 06:06 IST

During recurring **Chennai floods** (2015, 2023, recent monsoon events):

- Prolonged power outages leave **mobile phones and emergency devices unusable**
- **Cell towers and internet services fail** due to waterlogging and infrastructure damage
- Flood-trapped residents cannot send SOS messages or share their exact location
- Rescue teams struggle with missing or inaccurate location information, delaying evacuation
- Existing emergency systems depend on **telecom networks and signal**

Result:

As a result, critical rescue hours are lost during floods, directly increasing casualties due to the absence of last-mile emergency communication.

Proposed Solution



A No-Network Emergency Location System

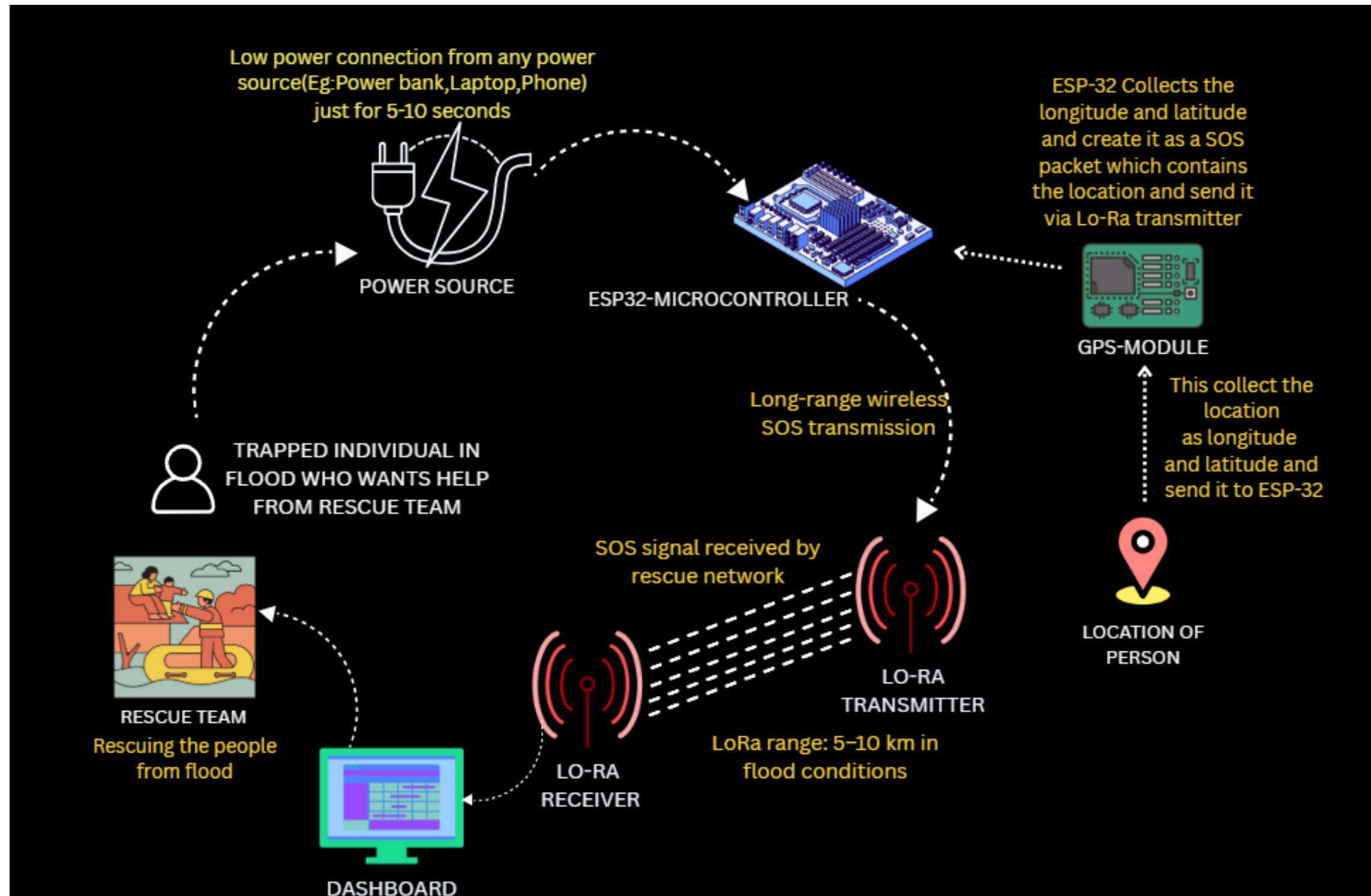
- Operates in complete network blackout conditions
- GPS-enabled to transmit exact latitude and longitude
- LoRa wireless communication for long-range SOS (5–10 km)
- **SOS messages can be received by any nearby LoRa rescue node or temporary gateway used by government, NGOs, or rescue teams.**

A portable LoRa gateway connected to a laptop forwards the message through other nodes to the rescue control room, allowing quick deployment in flood areas.

Outcome:

A last-resort emergency communication system designed to function during network collapse, ensuring victims can still transmit precise location data to rescuers.

Technical Description



The Trapped person is rescued by using a little amount of power(powerbank,mmobile,etc) to run esp32 and then the signal is transmitted by lora and then the location with date and time is recieved by lora reciever team within seconds

Technology used :

HARDWARE:

- ESP-32 Microcontroller
- LoRa Module: SX1278 (433MHz)
- GPS Module
- Led Display(16x2)
- Antenna: 433 MHz spring antenna

SOFTWARE:

- Arduino IDE(Microcontroller)
- HTML,CSS,Java Script
- flask API
- leaflet.js
- Python (pyserial)

Impact & Benefits



Impact

- Enables emergency communication during Chennai floods despite high-power and network failure
- Can reduce location identification time from hours to minutes by providing exact GPS location of trapped victims
- Significantly reduces preventable loss of life by restoring communication during the most critical post-flood rescue window.
- Supports government and disaster-response teams with reliable SOS data
- Creates a scalable disaster-communication model for other flood-prone regions in India

Benefits

- It requires low power(5V) just for few seconds.
- Independent of mobile networks and internet
- Long-range communication using low-power LoRa technology
- Single-action operation ensures usability even under panic, injury, or low-visibility flood conditions.
- Low-cost and rugged, suitable for mass deployment
- Can be pre-positioned in flood-prone zones for rapid use

Feasibility & Viability



Feasibility

- Uses readily available components in India (Esp32, Lo-Ra, Led-Display)
 - No complex infrastructure required at the user side
 - Proven technologies (GPS + LoRa) already used in real-world deployments
 - Low power requirement, easily met by manual energy generation
 - Core system functionality — power generation, GPS acquisition, and LoRa transmission — can be assembled and demonstrated within hackathon timelines.

Viability

- Low-cost design, suitable for large-scale deployment
 - Minimal maintenance due to battery-less operation
 - Compatible with existing LoRa gateways used by government agencies
 - Can be deployed in flood-prone urban and rural regions
 - Scalable for disaster management, NGOs, and public safety programs

Competitor Analysis



EXISTING SOLUTIONS	WHAT IT DOES	Power / Network Dependency	Limitations in Flood Scenarios
MOBILE PHONES (CALL/SMS)	Voice/SMS emergency communication	Battery + Cellular network	Fails when battery drains or network is down
Mobile SOS Apps	Shares GPS via internet	Battery + Internet	Useless without data connectivity
Satellite Messengers	SOS via satellite	Battery + Subscription	Expensive, limited access
Walkie-Talkies / VHF Radios	Local voice communication	Battery-powered	No GPS, short range
Our Solution Advantage	Sends emergency SOS + GPS location	Works without proper signal and network	Minimal limitation

- The proposed system is among the few solutions capable of operating during complete power and network blackout while still providing accurate GPS location at low cost.
- When networks fail and there is no signal, RESCUE-ZERO remains operational.

Cost Estimate



Component	Quantity	Price
ESP-32	1	Rs. 300
LoRa (SX1278)	1	Rs. 300
GPS Module	1	Rs. 240
Antenna (433MHz)	1	Rs. 100
Capacitor, Resistor	4	Rs. 30
Casing (3D Printing)	1	Rs. 300
	TOTAL COST	Rs. 1270

- Estimated Prototype Cost: ₹1,270 (Affordable & Scalable)
- **Cost can be reduced further in bulk manufacturing**
- 3D-printed casing can be replaced with injection-molded enclosure
- Significantly cheaper than satellite phones or battery-based emergency devices